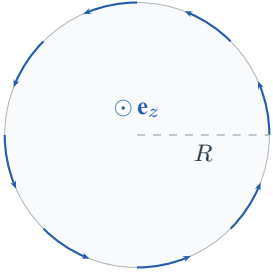
M1-41 圆盘上的单位法向旋度



C 逆时针；$\hat{\mathbf{n}} = +\mathbf{e}_z$

$$\mathbf{A} = (-y/2, x/2, 0)$$

$|\mathbf{A}| = R/2$，沿圆周切向

$$\oint_C \mathbf{A}\cdot d\mathbf{l} = (R/2)(2\pi R)$$

$$\nabla \times \mathbf{A} = \mathbf{e}_z$$

环量与圆盘面积均为 πR^2。